

Exploring key categories of social perception and moral responsibility of AI-based agents at work: Findings from a case study in an industrial setting

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ABSTRACT

The increasing adoption of smart and interactive technologies, based on Artificial Intelligence (AI), in business and society transforms humans' perception of these technologies towards social actors. However, since there are also reservations of humans to interact with artificial agents, it is fundamental to explore the social perception of this new technology and the implications thereof. Social perception governs social behavior and subsequently affects interactions with AI-agents. Moral responsibility plays a pivotal role in functioning societies and should thus be taken into account when implementing AI. This qualitative study examines workers in an industrial setting, who perform a decision-making task together with an AI-agent possessing no visual human-like characteristics. Based on the model of Responsibility in Hybrid Societies we show that workers assign capacities of social perception and moral responsibility to their AI-based counterpart. We present first evidence that theoretically derived models of social perception and moral responsibility are applicable in industrial settings.

CCS CONCEPTS

• **Human-centered computing** → Human computer interaction (HCI); • **Applied computing** → Law, social and behavioral sciences; • **Computing methodologies** → Artificial intelligence;

KEYWORDS

Human-AI-Collaboration, SPRS social perception of robots scale, responsibility, modern work, qualitative analysis

ACM Reference Format:

Sarah Mandl, Maria Kobert, Maximilian Bretschneider, Frank Asbrock, Bertolt Meyer, Anja Strobel, and Thomas Süße. 2023. Exploring key categories of social perception and moral responsibility of AI-based agents at work: Findings from a case study in an industrial setting. In *Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems (CHI EA '23)*, April 23–28, 2023, Hamburg, Germany. ACM, New York, NY, USA, 6 pages. <https://doi.org/10.1145/3544549.3585906>

1 INTRODUCTION

With technologization moving forward, workspaces are currently being redesigned: One major change is the application of new, increasingly smart technologies. For example, industrial robots, often equipped with Artificial Intelligence (AI), are used in areas that have usually been occupied by humans, and thus are brought into play as social actors. In the present study, we examine the collaboration between shop floor workers and an image recognition AI-agent based on a machine-learning algorithm in a branch of an

automotive supplier in Germany. The AI-agent has the appearance of an industrial machine, consisting of a black rectangular box with cameras inside, a conveyor belt, and a sorter. The joint task of the human-AI team is to visually inspect used material for quality and to decide collaboratively whether it can be reused or has to be disposed of. To do so, the workers supply the AI-agent with the material using a conveyor belt. The AI-agent inspects the material and sorts it into boxes according to the material's quality. In the end, the human verifies the AI-agent's decision. In case the AI-agent has made a mistake, the worker provides it with feedback in the form of labeled image data, such that the AI-agent improves over time.

Our aim is to investigate how the human counterparts perceive this new artificial actor in their work environment in terms of social dimensions and attributed responsibility by drawing from interview data.

The dramatically increased use of digital technologies like robots and AI-systems in workspaces, which are becoming more interactive and intelligent, over the last years [2, 22, 27], also received some critique. Often discussed issues are, for example, reduced direct contact between humans, decreased social interaction [6, 23], monotonous working conditions [1], and the perceived threat of large-scale job losses [9]. However, since changes are rarely stoppable, it is essential to provide insights into how these technologies are perceived and subsequently accepted by users, how these evaluations might change over time, and which adaptations could improve initially negative attitudes.

Social perception. One aspect that could contribute to the acceptance of new technologies is their social perception. Social perception influences interaction in societies [5]. In 1994, Nass et al. [23] proposed that artificial agents are perceived as social actors in their own right. This paradigm, Computers are Social Actors (CaSA), is substantiated by an array of studies, showing that computers are perceived as social actors (not to be confused with anthropomorphism), even though this attribution is negated as inappropriate by users [23]. This paradigm has implications for current research on Human-Robot Interaction (HRI) since artificial actors are often perceived as social actors by their users [6, 25]. This is further demonstrated since robots in working environments are perceived as social entities and are attributed positive and negative humanlike characteristics [24]. Traditionally, social groups are categorized along two (competence and warmth, e.g., [4, 5, 8]) or three (competence, sociability, morality, e.g. [13, 14]) dimensions that affect emotions and behavior toward these social groups. For example, groups perceived as low in competence and high in warmth evoke feelings of pity and sympathy, whereas groups perceived as high in competence and low in warmth evoke feelings of envy and jealousy [8]. Prior research showed that such social dimensions were indeed attributed to robots [3, 17]. However, these dimensions differed from the traditional human dimensions insofar as that factorial analyses revealed three distinct facets of social perception: anthropomorphism, sociability/morality, and activity/cooperation [16]. Anthropomorphism alludes to the perceived human-likeness of an artificial agent, that is, whether it moves smoothly or rigidly, or whether it appears to have a consciousness or not [16]. Sociability/morality refers to the perception of the artificial agent as a social interaction partner, that is, for example, whether it is perceived as moral or immoral, or as likeable or unlikeable [16]. As mentioned,

robots in work settings can be perceived as social entities, and thus, it should be taken into account whether these evaluations are positive or negative [18]. Finally, activity/cooperation refers to aspects important in the actual interaction in work settings, for example, whether the artificial agent is perceived as fair or unfair, or as diligent or lazy [16]. Often overlooked, this dimension could give important insights on how users evaluate the helpfulness of the robot while working with it. Furthermore, competence should be considered when humans interact with artificial agents: Whether an artificial agent is perceived as competent or incompetent has tremendous impact on the overall evaluation, for example in industrial settings. Competence was attributed to robots in previous works: industrial robots were attributed more competence than social robots [3, 17], and a humanlike robot was perceived as more competent than a zoomorphic one [15]. McKee et al. [18] presented evidence that people preferred incompetent AI-agents over competent ones, which could point to differences depending on application areas.

Responsibility in Hybrid Societies. To integrate new technologies seamlessly into human societies, and thus also into workspaces, responsibility plays a pivotal role [19]. Responsibility in this context is understood as both legal and moral responsibility and is attributed to different actors of Hybrid Societies, that is, societies consisting of humans and artificial entities, to a varying extent [19]. Responsibility in a legal sense includes the capacity to act (the ability to act consciously), legal capacity (the ability to be the bearer of rights and obligations), autonomy (with higher autonomy corresponding with higher ability to accept the duty to assume responsibility), liability (practical responsibility in a legal sense), and explainability (the understanding of why an actor acts as it does, and thus, to understand why legal conditions are created in a certain way)[19]. Moral responsibility from a psychological perspective includes explainability, moral agency, and trust [19]. Explainability in this context should be understood as a precursor for trust: only if users – including lay users – can understand why a given actor acts as it does, they can trust this actor. Moral agency refers to a plethora of capacities, e.g., thought, communication, emotion, memory, morality, or self-control [19] which need to be present in an agent to act morally responsible. It is thus a capacity that presupposes aspects of autonomy and intentionality [10]. Trust is, in this context, understood as the trust placed in an agent to fulfill its purpose without causing harm and with low failure-rates [19]. It is thus a factor greatly influencing acceptance and subsequent use of artificial agents [19] and should be evaluated cautiously: Trust placed in an artificial agent should be balanced since over- and under-reliance have adverse effects on the evaluation and acceptance of a given agent [11, 19]. To be fully responsible, these capacities need to be met. For now, only human beings are capable of full responsibility [19]. However, this does not mean that certain capacities could not be attributed to artificial agents in the sense that these actors *appear* to possess the capacities [20].

Consequently, the present study is concerned with the attribution of social perception dimensions (anthropomorphism, sociability/morality, activity/cooperation) and moral responsibility (explainability, moral agency, trust) towards an AI-agent deployed in a remanufacturing process in a large automotive supplier in

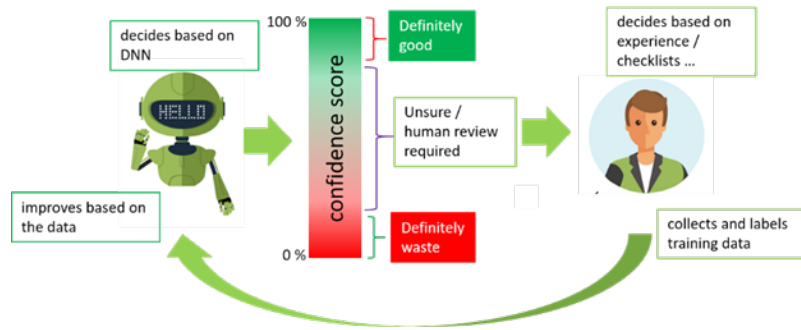


Figure 1: Allocation of tasks within the human-AI Team

Germany. We are investigating whether aforementioned social perception dimensions and capacities of moral responsibility can be found in workers' evaluation of their experience while working within a human-AI team.

2 CASE STUDY DESCRIPTION

We performed a case study at a branch of a large automotive supplier in Germany. The company's business model has been focused on remanufacturing used car clutches since the 1970s. At the company, we examined a distinct area of the remanufacturing process, namely, the quality control of used compression springs that are parts of the clutches. Shop floor workers had manually performed visual quality control of the springs until about two years ago. The company then introduced an AI-agent, which has supported the employees since.

The participants of the in-depth study are six employees who are directly or indirectly involved with the AI-agent. Three of them are shop floor workers, collaborating directly with the AI-agent on a daily basis. One is the supervisor of the company's branch to which the shop floor workers belong, and one is their direct foreman. Additionally, we interviewed the quality manager, who is responsible for defining the quality criteria for the springs. All participants are male, with an average age of 45 years. Some of the workers have been instructed by the AI-agent's developer before they started working with the machine.

The AI-agent is based on a deep neural network and is able to learn and perform a visual recognition task. It is put into operation with the capability of handling simple and obvious cases of a task independently and handing over complicated cases to the employees. The AI-agent is trained repeatedly by providing an appropriate amount of training data. Thus, the number of cases which are handed over to the employees is decreasing over time because the AI-agent is continuously improving. The AI-agent has the physical appearance of an industrial machine, consisting of a screen, a conveyer belt, a rectangular black machine housing, and a sorter. The compression springs are manually placed onto the conveyer belt and are then transported inside the machine housing. There, four cameras are installed that produce images of the compression spring from different perspectives. Based on these images, the springs are evaluated by the AI algorithm. The images and the AI-agent's evaluation results are displayed on the screen. Positive evaluations are outlined in green, and negative evaluations in red. There is also a number displayed, which can range from 000 to 099,

where 000 stands for "spring can be wasted" and 099 for "spring is perfectly ok". In the end, the springs are automatically sorted into different boxes. The sorter can handle up to five different evaluation ranges.

The human-AI team's joint task is to visually inspect used compression springs for quality and, thus, to decide collaboratively whether the spring can be reused or should be disposed of (see Figure 1). Therefore, firstly, the worker supplies the machine with the spring by putting it onto the conveyer belt. After that, the AI-agent classifies the quality of the spring and sorts the springs into boxes depending on its confidence; for example, definitely waste, not sure, and definitely reusable. The workers then recheck the boxes containing the springs that the AI-agent was uncertain about and, finally, sort these springs into good and bad ones as well. They subsequently collect the springs the AI-agent was either unsure about or those with defects that the AI-agent is unable to recognize yet. Images are then taken of those springs by the AI-agent, which are then saved and labelled at a later date by the workers. These images represent the training data. The workers provide valuable impulses for the targeted enrichment of the training data by constantly comparing their own evaluations with those of the AI-agent. The actual training process, however, is executed by the AI's developer.

3 METHODS

We gathered empirical data by conducting semi-structured interviews with the employees, who have already been characterized in section 2.1. Consequently, we developed an interview guideline. The interview guideline contained simple questions focusing on the participant's personal impressions and experiences of the collaboration with the AI-agent, without actively hinting at above mentioned facets. A few sample questions are: "What are particular challenges you experience when working with the AI-agent?", "What do you like or dislike when working with the AI-agent?", "How does working with the AI-agent differ from working with other machines?", "What changes have you experienced since the AI-agent was introduced?"

The interviews took place in the summer of 2021. Each of the interviews lasted between 30 and 45 minutes and was audio-recorded. After conducting the interviews, we transcribed the audio files and subsequently performed a qualitative content analysis [21]. We

Table 1: Social perception and moral responsibility dimensions attributed to an industrial AI-agent

	Example Quotes	Frequency	
Social perception			
Anthropomorphism	„Because, after all, the AI had to learn like a normal human.”	15	
Sociability/Morality	„Nah, it runs with us as Ferdinand or AiZubi (AiTrainee).”		
	Sociability Morality „But ultimately it does not have to be that, eh, that the machine is burdened with something like this.”	5	-
Activity/Cooperation	Activity „The [AI] just doesn’t take a break, it’s just, it can run twenty-four hours a day.”	2	3
	Cooperation „The finished springs, they’re fine, very good, we then put those in, tell the AI „the images you see now are evaluated as fine, so must be evaluated to be fine”, and then we push [the springs] through”.		
Competence	„What the AI can already do for sure is the trash box, where 100% of the springs are not alright, it can already do that very well.” „Of course, you have to keep rechecking because the AI is not 100% accurate yet and um, but bit by bit it gets better and better.”	23	
Moral responsibility			
Trust	„So we thought about how we could build up the system at least to the point where we could say that we trust it.” „The machine does something and says afterwards, so when we’re completely finished, eh, “that’s okay, so that’s a bad one, that’s a good one”. I can rely on it.”	9	
Explainability	„So on the screen you actually see every, really every camera perspective. And then you look, if you, it’s a relatively stupid [indiscernible], stupid work there, to put the parts on it. You look at it, um, because every box is behind, behind, underlaid then. Red means I see the error, green means I see no error. Sometimes it’s quite interesting to see what the machine is doing, how it’s evaluating.” „It’s black, exactly, eh, you can’t see much either, because opening the lid and stuff like that isn’t possible, eh, they know that the cameras are inside, nah, but otherwise that’s of course first of all, the big unknown object, that’s standing there, nah;”	4	
Moral Agency	„So that [the AI] then just does everything autonomously anyway and you just end up packing stuff”. „Yes, at the moment we still have to think for ourselves anyway, but that should change at some point in the future. That the machine then takes over the largest part of the thinking process.”	6	

applied the software MAXQDA Version 2020 [26] for the data analysis. We employed an iterative process during our analysis. Firstly, we analyzed and coded the data independently from each other. After that, we discussed our interpretations, looked for relationships and patterns, constructed categories and grouped codes. In the process, we applied a mix of deductive and inductive coding [12] by assigning the codes to concepts, which were based on either our theoretical framework or insights that emerged during data collection and analysis. Additionally, feedback loops took place with some participants of the case company's project management. We refined the codes iteratively until consensus was reached among all researchers participating.

4 FINDINGS

In our transcripts, we identified four dimensions of social perception and three dimensions of moral responsibility, which correspond to

the ones we already introduced above. We present representative example quotes for each dimension in Table 1. We found no mentions of morality in the interviews, which is unsurprising given the context of an industrial machine with no human-like representation. The dimension of moral agency is mainly represented by the aspect of autonomy.

5 DISCUSSION

With the present study, we made an explorative effort to investigate whether constructs of social perception, namely anthropomorphism, sociability/morality, activity/cooperation [16], and competence [3, 8, 17, 18] as well as moral responsibility in Hybrid Societies, namely trust, explainability, and moral agency [19] previously derived theoretically can be applied to AI in industrial work environments. To this end, we analyzed interviews with six workers directly or indirectly involved with an AI-agent in an industrial

work setting. The workers took part in a collaborative task in such a way that one clearly defined work step, the visual inspection of a spring in terms of its reusability, was taken over jointly by the AI-agent and the worker.

We looked for signs of workers' social attributions towards the agent, i.e., for anthropomorphism, sociability/morality, activity/cooperation, and competence and attributions of moral responsibility, namely trust, explainability, and moral agency. Despite the industrial setting and the physical appearance of the AI-agent as a rectangular box, we found several statements that referred to social dimensions and moral responsibility of the AI-agent. This indicates that the AI-agent, even though a non-humanlike machine, was seen, at least to a certain extent, as a social actor, lending further evidence to the CaSA paradigm [23] and Demir et al.'s intuition, that machines can be seen as co-workers [7]. We highlight that these references were stated without being prompted by our questions, indicating the relevance of above mentioned constructs in business applications of AI-agents. In the following, we shortly discuss the attributions mentioned above.

Anthropomorphism. The AI-agent in question had the physical appearance of a machine and was thus not designed to be reminiscent of a human. It is all the more remarkable that, even though the physical appearance is not indicating so, workers' statements very clearly showed that they have anthropomorphized the AI-agent, that is, attributed humanlike capacities such as a name.

Sociability/morality. Across six interviews, only few statements referred to the AI's perceived sociability and none to morality. With respect to the application of the AI in the described industrial setting, it was not surprising that sociability/morality would play a subordinate role. However, this dimension should not be interpreted as irrelevant. As McKee et al. [18] stated, humans interacting with AI-agents preferred warm over cold agents, so the importance of sociability/morality and/or warmth could be different in settings where social interaction between the human and the AI is more essential, such as care or education.

Activity/cooperation. In terms of activity/cooperation, workers referred to the cooperative nature of the human-AI interaction in this specific setting, indicating that they rudimentarily communicated with it as can be seen in this statement "[we] tell the AI 'the images you see now are evaluated as fine, so must be evaluated to be fine'". Furthermore, we found that the AI-agents' level of activity, i.e., it does not need breaks, was also emphasized by the workers.

Competence. Unsurprisingly, statements referring to the AI-agents' competence were easy to find. The interviewees mentioned both, that the AI was (or will be) competent in what it does, that is, a distinct part in the work process, and that, at the moment, it was still not foolproof and needed supervision. This emphasizes that, especially for an AI-agent with a distinct task, competence played a major role in its perception by humans.

Moral responsibility. With regard to moral responsibility, all three capacities, that is, trust, explainability, and moral agency, were represented in the workers' statements. It is then not surprising that, given the nature of the task the AI-agent is deployed in, especially trust and explainability were present in the quotes. Some workers indicated that, even though they used the AI-agent for about two years now, they do not really understand what the agent does. This indicates that, although theoretically explainability

should be considered as a major aspect of moral responsibility [19], at least in this specific use-case, it is not always given. Furthermore, workers referred to the trust they placed (or did not place) in the AI-agent. Interestingly, we found statements referring to the autonomy of the AI-agent. Autonomy plays an important role in HCI in different contexts, e.g., in legal responsibility [19]. However, it is also considered as a capacity necessary for moral agency [19]. While moral agency as a capacity is only applicable to humans so far [19], our findings indicate that sub facets of moral agency, such as trust, explainability, or, possibly, autonomy, can also be attributed to AI-agents. However, whether autonomy in this context is perceived as a precursor for moral agency needs further research.

For the successful implementation of AI-agents in work settings and, consequentially, in private settings, aspects of social perception and moral responsibility seem to be relevant for smooth integration. Until now, these aspects were considered based on theoretical considerations. The analysis of the qualitative data, however, shows first evidence that those aspects play a considerable role in the adoption of an AI-agent in an industrial setting and should thus be considered while new technologies are implemented. Follow-up studies should take on the endeavor of extending these first results and also provide insights into different types of AI-agents in diverse settings, e.g., implemented in robots.

6 LIMITATIONS

Our study has certain limitations. Primarily, our sample consisted of individuals from only one German company representing a specific use case. So, the results may not represent a larger population of employees working with AI-agents. Secondly, gathered opinions, experiences, and interpretations are limited due to personal experience and knowledge. Finally, despite the utmost methodological cautiousness, the results may unintentionally be biased by the researchers during data collection or categorization. These limitations are typical for qualitative studies; the results are neither fully representative nor generalizable and are not simply transferable to different cultural contexts.

7 CONCLUSION AND FUTURE WORK

Despite the limitations mentioned above, our study provides first relevant insights into the human parts' social perception in human-AI systems related to a specific case which we described in detail throughout the manuscript. The proliferation of AI systems is likely to fundamentally change job characteristics and employment patterns. For instance, using AI systems will change how workers perceive their job and their motivation to cooperate with such systems. Both aspects are closely linked to the social perception of AI systems in that way that negative attributions might lead to unwanted side effects. In contrast, positive social perception offers the opportunity to promote technology acceptance and workers' motivation in the long run. The focus on the social perception of such technologies is highly relevant to ensure appropriate design and usage. However, research on these specific aspects is still scarce and further clarification is needed.

Consequently, we addressed this issue in an exploratory way. We presented first evidence that the theoretically derived capacities of social perception and moral responsibility can indeed be found in

statements made by humans who are working with an industrial AI-agent. More precisely, humans working with industrial AI-agents attribute capacities of social perception and moral responsibility to their mechanical counterparts. Therefore, our findings can be seen as a first step to validate the models for social perception [16] and moral responsibility [19] within industrial settings. As future work, it is planned to extend the present industrial context database to generalize the results and to conduct qualitative and quantitative studies in different working and social environments or settings to validate and generalize our findings.

ACKNOWLEDGMENTS

Funded by the Deutsche Forschungsgemeinschaft (DFG, German Research Foundation) – Project-ID 416228727-SFB 1410.

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